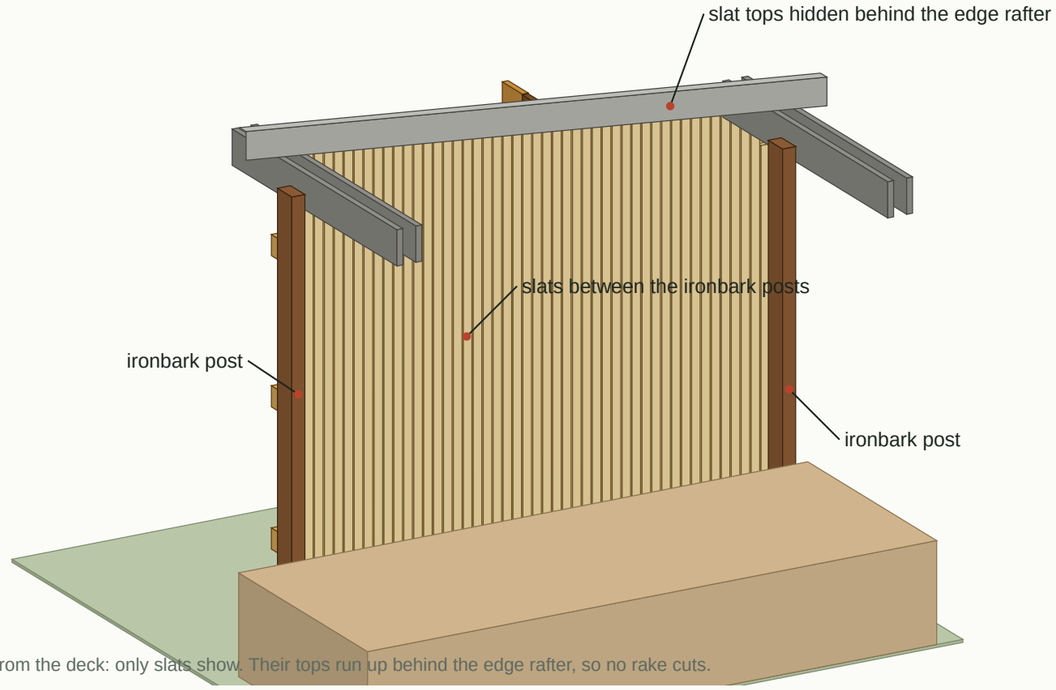
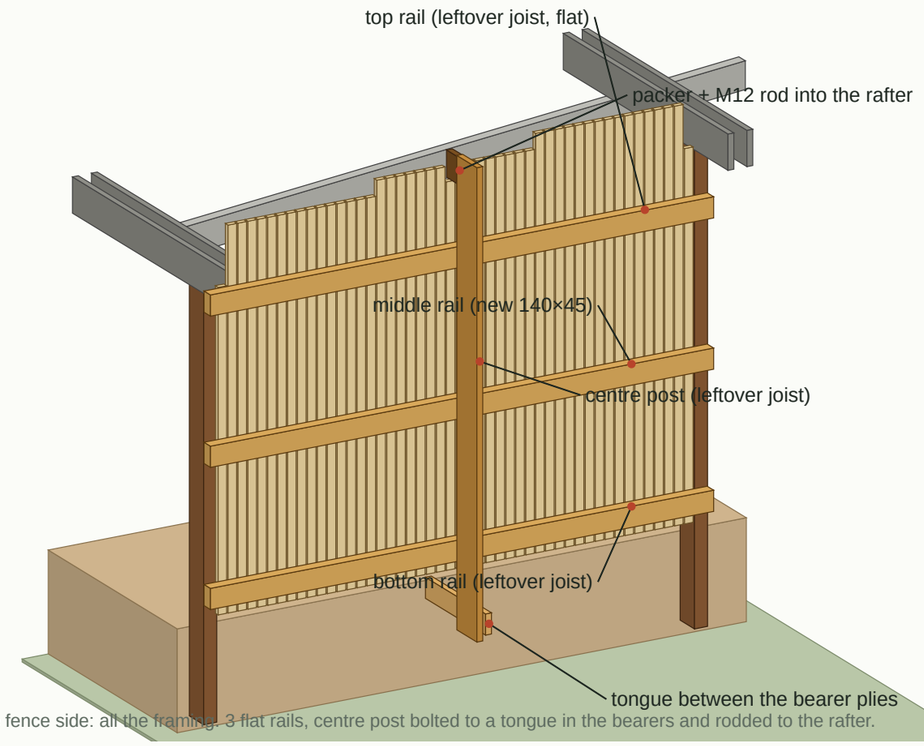


From the deck: only slats show. Their tops run up behind the edge rafter, so no rake cuts.



From the fence side: all the framing. 3 flat rails, centre post bolted to a tongue in the bearers and rodded to the rafter.

AT A GLANCE

Clear span between ironbark posts	3 660
Rails (flat, fence face of posts): bottom / middle / top	100–240 / 1 040–1 180 / 2 040–2 180
Centre post, 1 825 from the studio-end post	3136 long
Slats	48 @ 65, gaps 11
Slat lengths A / B / C	2775 / 2670 / 2565
Slat bottom above deck	20

Why the centre post: a 140×45 rail laid flat can't span the full 3.66 m in a big wind. Halving the span to 1.83 m brings it to about 12 MPa against an MGP10 limit of about 16, with about 8 mm of bow.

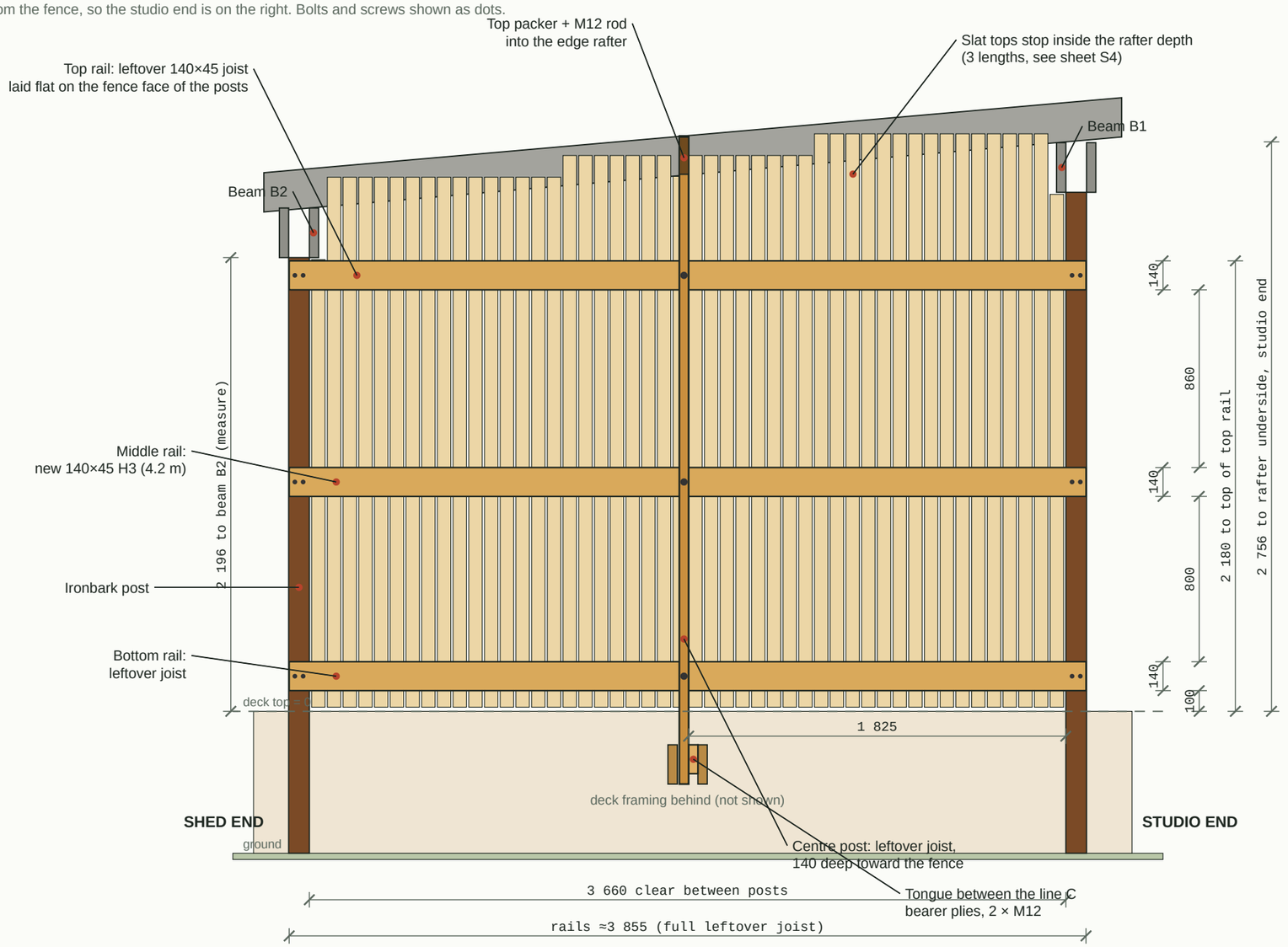
MATERIALS AND HARDWARE

ITEM	SIZE	QTY	USE
Leftover deck joist, H3	140×45 × ≈3 855	3	top rail, bottom rail, centre post (+ 420 tongue from the post offcut)
New H3 MGP10	140×45 × 4.2 m	1	middle rail, trim to match the others
H3 pine screening slats	65×19 × 3.0 m	50	48 + 2 spare, cut to 3 lengths (S4)
Offcuts 70×35	≈200	4	top packer (2 flat = 70) and tongue packers
Hex batten screws, Type 17, galv	14g × 100	12	rails to ironbark posts, 2 per crossing, pre-drill
Galv bolts, nuts, 2 washers	M12 × 200	3	each rail to the centre post
Galv bolts, nuts, 2 washers	M12 × 120	2	centre post to tongue
Galv bolts, nuts, 2 washers	M12 × 220	2	tongue through both bearer plies
Galv threaded rod, nuts, washers	M12 × 300	1	post → packer → rafter
Stainless or galv nails / screws	50 nail or 8g×40	≈300	2 per slat per rail
End-grain sealer		1	all cut ends of H3

Check on site before cutting: clear span, rafter underside at both ends, beam B2 underside, slat-to-rafter gap. Structure to be confirmed by Ollie Harris before building.

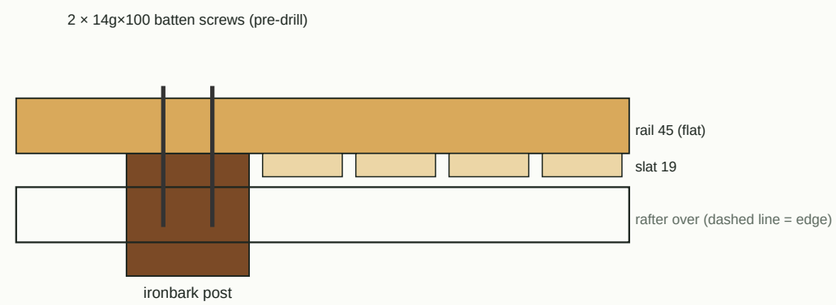
ELEVATION FROM THE FENCE SIDE (1:25)

Shown from the fence, so the studio end is on the right. Bolts and screws shown as dots.



PLAN AT A RAIL, IRONBARK POST (1:5)

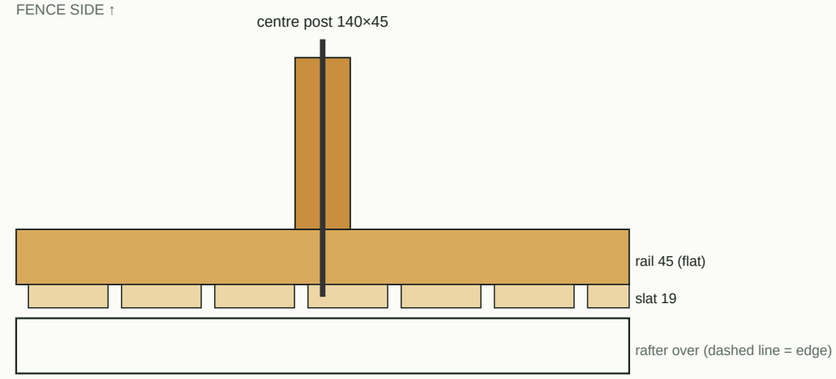
FENCE SIDE ↑



DECK SIDE ↓

PLAN AT A RAIL, CENTRE POST (1:5)

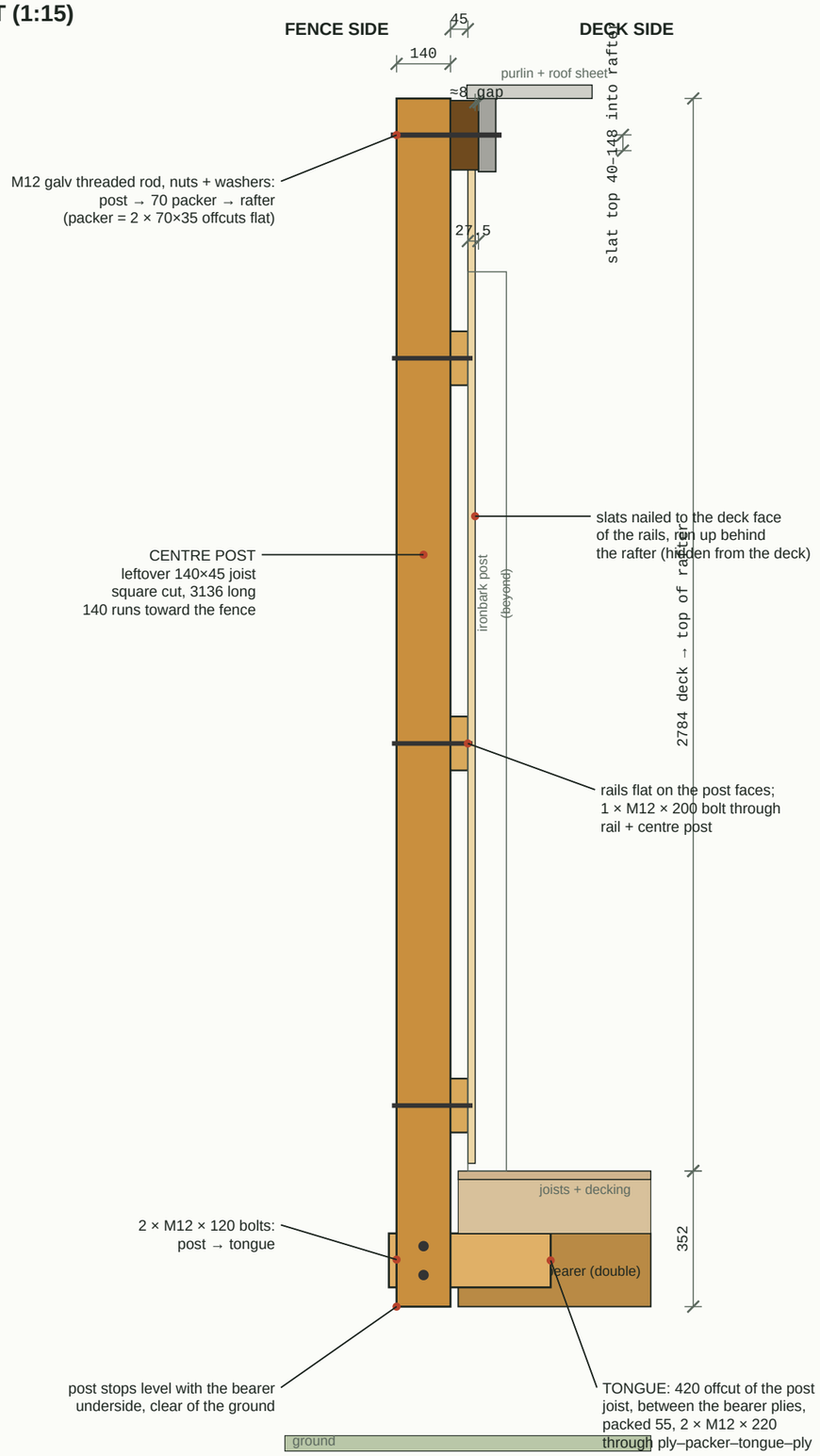
FENCE SIDE ↑



DECK SIDE ↓

Check on site before cutting: clear span, rafter underside at both ends, beam B2 underside, slat-to-rafter gap. Structure to be confirmed by Ollie Harris before building.

SECTION THROUGH THE CENTRE POST (1:15)



FIXING SCHEDULE

JOINT	FIXING
Rail → ironbark post	2 × 14g × 100 hex batten screws per crossing, pre-drilled 5 mm (ironbark splits)
Rail → centre post	1 × M12 × 200 bolt per rail, through the 45 rail and 140 post, washers both sides
Centre post → tongue	2 × M12 × 120 bolts, 75 apart vertically
Tongue → line C bearer	2 × M12 × 220 bolts through ply, 55 packer, tongue, ply
Centre post top → rafter	1 × M12 threaded rod through post, 70 packer and rafter, washers and nuts both ends
Slat → rail	2 × 50 stainless or galv nails (or 8g × 40 stainless screws) per rail, 15 in from the edges

HOW THE LOADS GO

- Wind on the slats goes into the 3 rails.
- Each rail spans 1.83 m, from an ironbark post to the centre post and on to the other ironbark post.
- The centre post spans about 2.8 m, from the tongue in the bearers up to the rafter, and hands its load to both.
- Slat tops are unfixed and hang up to 615 above the top rail. That's fine for 65 × 19 in strength and stiffness.

The ≈8 mm gap between the slat face and the side of the rafter is from the model. Hold a slat up and check it before you cut. If a slat rubs, flip it or take a plane pass off the back.

Optional: to stop the slat tops rattling, put one 8g screw through each slat into the side of the rafter, with an 8 mm packer strip behind. Screw from the fence side so it can't be seen from the deck.

SLAT CUTTING SCHEDULE

GROUP	SLAT NO. (FROM THE STUDIO END)	CUT LENGTH	TOP ABOVE DECK	NOTE
A	1–16	2775	2795	studio end → shed end: tops sit 40–145 into the rafter
B	17–32	2670	2690	studio end → shed end: tops sit 41–147 into the rafter
C	33–48	2565	2585	studio end → shed end: tops sit 43–148 into the rafter
—	1	2482	2502	stops 10 under beam B1
—	25	2569	2589	stops 10 under the top packer
—	48	2166	2186	stops 10 under beam B2

The rafter drops about 316 across the span, so one length can't stay hidden behind it the whole way. The three groups each step down 105, which keeps every top 40–148 up inside the 190 rafter: hidden from the deck and clear of the purlins.

CHECKS BEFORE YOU CALL IT DONE

- No slat top shows below the rafter from a seated position on the deck.
- No slat touches the purlins or the roof sheet.
- Rails don't touch beam B2 (top rail 2 180 vs beam ≈2 196).
- Centre post plumb, nuts tight, rod snug.
- All cut ends sealed.

> Wind assumption: about 1.1 kPa ultimate on a screen that's 80% solid, with MGP10 H3 framing. The joints at the tongue and the rafter take the most load. Ollie to confirm before building.

BUILD ORDER

1. **Measure first.** Check the clear span between the posts (3 660), the rafter underside at both ends (≈2 756 studio, ≈2 436 shed, above the deck), beam B2's underside (≈2 196) and the slat-to-rafter gap. If any figure is more than 20 off, adjust the slat lengths in the table.
2. **Mark the rail heights** on the fence face of both ironbark posts: 100, 1 040 and 2 040 to the underside of each rail. Use a level or laser off the deck, not the rafter.
3. **Cut the centre post joist:** post 3136 square both ends, and the 420 tongue from the offcut. Seal the ends.
4. **Fit the tongue.** Slide it between the two line C bearer plies at the fence-side end, against the studio-side ply. Pack the 55 gap with 70×35 offcuts, then bolt with 2 × M12 × 220. It projects 180 past the bearer ends.
5. **Stand the centre post.** Bolt it to the fence-side end of the tongue with its deck-side face exactly one rail thickness (45) back from the fence face of the ironbark posts, so the rails sit flat across all three. Clamp it, plumb it both ways, then drill and fit the 2 × M12 × 120.
6. **Fix the top.** Pack between the post and the rafter with 2 × 70×35 offcuts laid flat (70). Drill through post, packer and rafter and fit the M12 rod. Snug it, don't crush it.
7. **Hang the rails.** Bottom first, then middle, then top. Clamp each one on its marks, check it's level, pre-drill and fix 2 batten screws into each ironbark post, then drill and bolt through the centre post.
8. **String-line check:** the deck face of the 3 rails should be dead straight. Pack behind at the centre post if needed before the slats go on.
9. **Cut the slats** to the 3 lengths, plus the 3 specials (table). Seal the cut ends. Each length is one stop-block setting.
10. **Slat set-out:** start at the studio-end post with an 11 mm spacer, bottoms on a 20 packer off the deck. Nail 2 per rail. Check plumb every 6 slats and fudge the gaps slightly if needed so the last gap at the shed-end post matches.
11. **Finish:** oil or stain to match the deck. Ends sealed.